1. Project Overview

This project focuses on analyzing customer shopping behavior using transactional data from 3,900 purchases across multiple product categories. The objective is to derive insights on spending patterns, customer segments, product preferences, and subscription behavior to inform strategic business decisions.[1]

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key feature groups:
  - Customer demographics: Age, Gender, Location, Subscription Status.[1]
  - Purchase details: Item Purchased, Category, Purchase Amount, Season, Size, Color.[1]
  - Shopping behavior: Discount Applied, Promo Code Used, Previous Purchases, Purchase Frequency, Review Rating, Shipping Type.[1]
- Missing data: 37 missing values in the Review Rating field.[1]

3. Exploratory Data Analysis (Python)

We performed data preparation and exploratory analysis in Python using pandas.[1]

- Data loading: Imported the raw dataset into a pandas DataFrame.
- Initial exploration: Used `df.info()` to inspect structure and `df.describe()` to review summary statistics.[1]
- Missing data handling: Identified nulls and imputed missing Review Rating values using the median rating within each product category.[1]
- Column standardization: Renamed columns to snake_case to improve readability and documentation.[1]
- Feature engineering:
  - Created an `age_group` feature by binning customer ages.[1]
  - Derived `purchase_frequency_days` based on purchase dates.[1]

- Data consistency checks: Assessed redundancy between `discount_applied` and `promo_code_used`, and removed `promo_code_used` after confirming overlap.[1]
- Database integration: Connected Python to PostgreSQL and loaded the cleaned dataset for SQL-based analysis.[1]

4. SQL Analysis (Business Questions)

We used PostgreSQL to execute structured queries and answer key business questions.[1]

1. Revenue by gender: Compared total revenue generated by male versus female customers.[1]

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-spending discount users: Identified customers who used discounts but still spent above the average purchase amount.[1]

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	6

3. Top 5 products by rating: Retrieved products with the highest average review ratings.[1]

	item_purchased text	average_product_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping type performance: Compared average purchase amounts between Standard and Express shipping.[1]

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers vs non-subscribers: Analyzed differences in average spend and total revenue by subscription status.[1]

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-dependent products: Found five products with the highest share of discounted purchases.[1]

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. Customer segmentation: Classified customers as New, Returning, or Loyal based on purchase history.[1]

	customer_segment text	number of customers bigint
1	returning	701
2	loyal	3116
3	new	83

8. Top 3 products per category: Listed the most frequently purchased products within each category.[1]

	item_rank bigint	category text	item_purchased text
1	1	Accessori...	Jewelry
2	2	Accessori...	Sunglasses
3	3	Accessori...	Belt
4	1	Clothing	Blouse
5	2	Clothing	Pants
6	3	Clothing	Shirt

9. Repeat buyers and subscriptions: Evaluated whether customers with more than five purchases are more likely to subscribe.[1]

	subscription_status text	repeated_customers bigint
1	No	2518
2	Yes	958

10. Revenue by age group: Calculated revenue contribution by age segment.[1]

	age_group text	total_revenue numeric
1	young adult	62143
2	middle-aged	59197
3	adult	55978
4	senior	55763

5. Dashboard in Power BI

An interactive Power BI dashboard was developed to visualize the key metrics and insights for stakeholders, enabling drill-down into customer segments, products, and time-based trends.[1]

6. Business Recommendations

Based on the analysis, the following data-driven actions are recommended:[1]

- Boost subscriptions: Promote clear, exclusive benefits to increase subscription uptake.[1]
- Strengthen loyalty programs: Reward high-frequency customers to transition them into the Loyal segment.[1]

- Optimize discount strategy: Review discount policies to balance volume growth with profit margins.[1]
- Enhance product positioning: Feature top-rated and best-selling products more prominently in campaigns.[1]
- Targeted marketing: Prioritize high-revenue age groups and customers who prefer express shipping for targeted promotions.[1]